OPMIG-05: Dynamic Routing & Bucket Management

Series: OPMIG | **Notebook:** 5 of 9 | **Created:** December 2025

OpenPipeline Migration Series | Notebook 5 of 9
Level: Intermediate
Estimated Time: 60 minutes

Learning Objectives

By completing this notebook, you will:

1. Master dynamic routing strategies for different migration scenarios
2. Design multi-tier bucket architectures for cost optimization
3. ⭐ **NEW:** Calculate ROI and TCO for bucket optimization strategies
4. ⭐ **NEW:** Implement 3-tier and 5-tier bucket governance models
5. ⭐ **NEW:** Apply compliance-driven retention strategies
6. Configure multi-pipeline processing patterns
7. Analyze bucket usage and identify optimization opportunities

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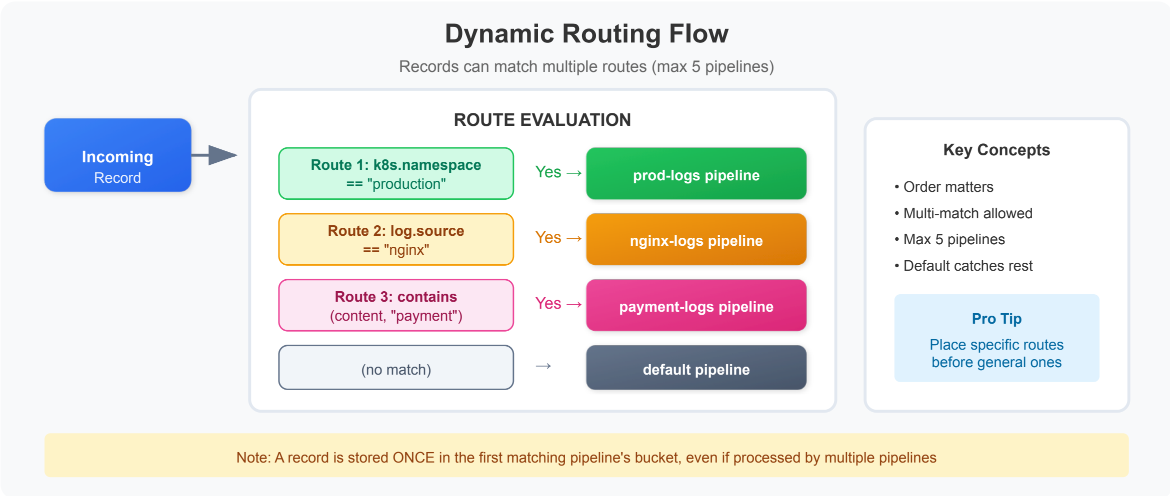
Understanding Dynamic Routing

Dynamic routing is the mechanism that directs incoming data to specific pipelines based on matching conditions.

How Routing Works



Key Routing Concepts

Concept	Description
Matching Condition	DQL expression that evaluates to true/false
Route Priority	Routes are evaluated in order (top to bottom)
Multi-match	One record can match multiple routes (max 5 pipelines)
Default Pipeline	Catches all records not matching any route
Exclusive Routing	Use specific conditions to prevent multi-match

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Routing Strategies

Strategy 1: Source-Based Routing

Route based on data source or ingestion method:

Condition	Target Pipeline	Use Case
`log.source == "nginx"`	nginx-logs	Web server logs
`log.source == "application"`	app-logs	Application logs
`dt.openpipeline.source == "otlp"`	otel-logs	OpenTelemetry
`dt.openpipeline.source == "generic"`	api-logs	API ingestion

Strategy 2: Environment-Based Routing

Route based on environment or namespace:

Condition	Target Pipeline	Bucket
`k8s.namespace.name == "production"`	prod-logs	default_logs (35d)
`k8s.namespace.name == "staging"`	staging-logs	dev_logs (7d)
`k8s.namespace.name == "development"`	dev-logs	dev_logs (7d)

Strategy 3: Content-Based Routing

Route based on log content patterns:

Condition	Target Pipeline	Purpose
`contains(content, "payment")`	payment-logs	Financial logs
`contains(content, "security")`	security-logs	Security events
`contains(content, "audit")`	audit-logs	Compliance logs

Strategy 4: Severity-Based Routing

Route based on log level for tiered retention:

Condition	Target Pipeline	Bucket
`loglevel == "ERROR"`	error-logs	error_logs (90d)
`loglevel == "WARN"`	warning-logs	default_logs (35d)
`loglevel == "DEBUG"`	debug-logs	short_retention (3d)

Strategy 5: Hybrid Routing

Combine multiple criteria for precise routing:

```
k8s.namespace.name == "production" AND log.source == "payment-service"
```

Grail Buckets Explained

Grail buckets are storage containers with configurable retention and access policies.

Default Buckets

Bucket	Data Type	Default Retention
`default_logs`	Log records	35 days
`default_spans`	Span/trace data	35 days
`default_bizevents`	Business events	35 days
`default_events`	Platform events	35 days

Custom Bucket Use Cases

Use Case	Bucket Strategy	Retention
Compliance/Audit	`audit_logs`	365+ days
Financial Data	`financial_logs`	90 days
Security Events	`security_logs`	180 days
Development Logs	`dev_logs`	7 days
Debug/Trace	`debug_logs`	3 days
High-Volume Noise	`ephemeral_logs`	1 day

Bucket Benefits

1. **Cost Optimization**: Shorter retention = lower storage costs
2. **Compliance**: Longer retention for audit requirements
3. **Performance**: Smaller buckets = faster queries
4. **Access Control**: Bucket-level permissions
5. **Data Isolation**: Separate buckets for different teams/purposes

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Bucket Routing Configuration

Configuring Bucket Routing in Pipeline

Each pipeline can route data to a specific bucket:

1. Open your pipeline in OpenPipeline settings
2. Go to **Storage** tab
3. Select target bucket from dropdown
4. Save pipeline

Bucket Routing Rules

Rule	Behavior
Pipeline bucket	Records go to pipeline's configured bucket
No bucket set	Records go to default bucket for data type
Multi-pipeline	Record stored once in first pipeline's bucket

> ⚠️ **Important:** A record is stored only ONCE, even if processed by multiple pipelines. It goes to the bucket of the FIRST matching pipeline.

Creating Custom Buckets

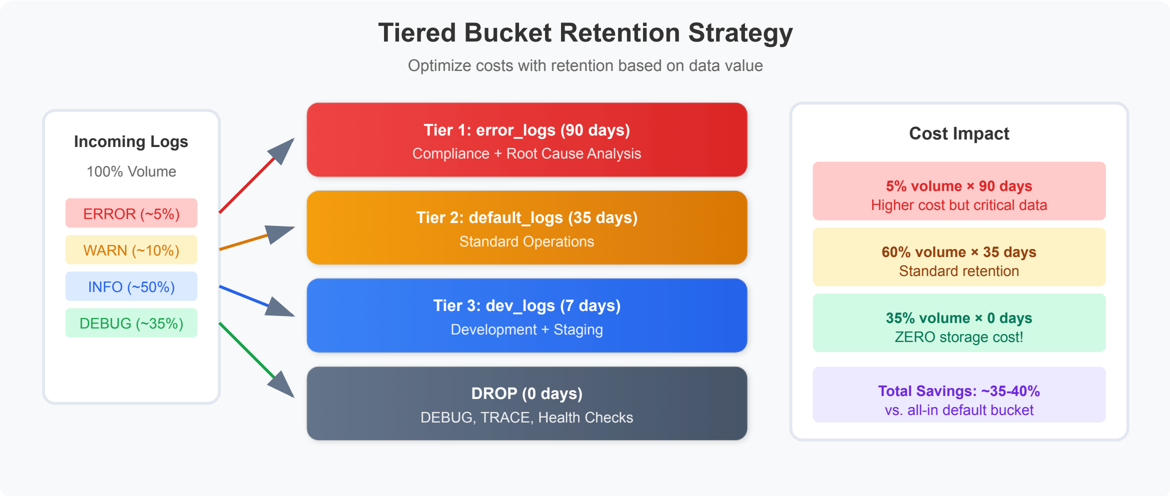
Custom buckets are created via Grail settings:

```
Settings → Data and storage → Grail buckets → + Create bucket
```

Configure:
- **Name**: Descriptive bucket name
- **Table**: Data type (logs, spans, etc.)
- **Retention**: Days to retain data
- **Description**: Purpose of bucket

Cost Optimization Patterns

Pattern 1: Tiered Retention by Severity



Pattern 2: Environment-Based Retention

Environment	Bucket	Retention	Cost Impact
Production	`prod_logs`	35 days	Standard
Staging	`staging_logs`	14 days	-60%
Development	`dev_logs`	7 days	-80%
Ephemeral/Test	`temp_logs`	3 days	-90%

Pattern 3: Drop + Sample

For extremely high-volume, low-value logs:

1. Drop 90% of health checks
2. Sample 10% to a short-retention bucket
3. Keep statistical visibility without full storage cost

Pattern 4: Metric Extraction + Drop

For logs needed only for metrics:

1. Extract metrics (with dimensions) from logs

2. Drop the raw logs after extraction
3. Metrics persist for 10 years, logs dropped immediately

Cost Optimization ROI Calculator ⭐ NEW

Understanding the financial impact of bucket strategies is critical for migration planning. This section provides detailed ROI calculations for common optimization patterns.

DDU (Davis Data Units) Pricing Model

OpenPipeline storage is measured in **Davis Data Units (DDUs)**:

Component	DDU Calculation
Log Ingestion	0.001 DDU per MB ingested
Storage (per day)	0.0001 DDU per MB per day retained
DQL Query	0.001 DDU per MB scanned

Typical DDU Pricing: ~$0.08 per DDU (varies by contract)

Cost Formula

```
Total Cost = (Ingestion Cost) + (Storage Cost) + (Query Cost)

Where:
  Ingestion Cost = Volume (MB/day) × 0.001 DDU/MB × Price/DDU
  Storage Cost   = Volume (MB/day) × Retention (days) × 0.0001 DDU/MB/day × Price/DDU
  Query Cost     = Query Volume (MB) × 0.001 DDU/MB × Price/DDU
```

Scenario 1: E-Commerce Platform - Tiered Retention

Current State (No Optimization):

```
Volume:           100 GB/day (100,000 MB/day)
Breakdown:        - ERROR:  5% (5,000 MB/day)
                  - WARN:  10% (10,000 MB/day)
                  - INFO:  50% (50,000 MB/day)
                  - DEBUG: 35% (35,000 MB/day)
Retention:        35 days (default_logs)
DDU Price:        $0.08 per DDU
```

Current Monthly Cost:

```
Ingestion:  100,000 MB/day × 30 days × 0.001 DDU/MB × $0.08 = $240/month
Storage:    100,000 MB/day × 35 days × 0.0001 DDU/MB/day × $0.08 =
$2,800/month
Query:      Estimate ~20% of stored data queried/month
            (100,000 × 30 × 0.2) × 0.001 × $0.08 = $48/month

TOTAL:      $3,088/month
```

Optimized State (3-Tier + Drop):

```
Tier 1 (ERROR):     5,000 MB/day → error_logs (90 days)
Tier 2 (WARN/INFO): 60,000 MB/day → default_logs (35 days)
Tier 3 (DEBUG):     35,000 MB/day → DROPPED (0 days)
```

Optimized Monthly Cost:

```
Ingestion (only non-dropped):
  65,000 MB/day × 30 days × 0.001 × $0.08 = $156/month

Storage:
  Tier 1: 5,000 × 90 × 0.0001 × $0.08 = $360/month
  Tier 2: 60,000 × 35 × 0.0001 × $0.08 = $1,680/month
  Total storage = $2,040/month

Query (20% of stored):
  (5,000 × 30 × 0.2 + 60,000 × 30 × 0.2) × 0.001 × $0.08 = $31.20/month

TOTAL:      $2,227.20/month
```

ROI Analysis:

```
Monthly Savings:     $3,088 - $2,227 = $861/month
Annual Savings:      $861 × 12 = $10,332/year
Savings Percentage:  27.9%
Implementation Time: 4-8 hours
Payback Period:      Immediate (operational change only)
```

Scenario 2: SaaS Platform - Multi-Environment

Current State:

```
Production:   50 GB/day × 35 days = $1,540/month
Staging:      30 GB/day × 35 days = $924/month
Development:  20 GB/day × 35 days = $616/month
TOTAL:        $3,080/month
```

Optimized State:

```
Production:   50 GB/day × 35 days = $1,540/month (no change)
Staging:      30 GB/day × 14 days = $369.60/month (60% reduction)
Development:  20 GB/day × 7 days  = $154.40/month (75% reduction)
TOTAL:        $2,064/month
```

ROI Analysis:

```
Monthly Savings:     $1,016/month
Annual Savings:      $12,192/year
Savings Percentage:  33%
```

Scenario 3: Global Retailer - Compliance + Cost

Requirements:
- Payment logs: PCI-DSS requires 90 days
- Audit logs: SOC 2 requires 365 days
- Application logs: 35 days standard
- Debug logs: Drop immediately

Volume Breakdown:

```
Payment logs:  10 GB/day (10% of total)
Audit logs:    5 GB/day  (5% of total)
App logs:      60 GB/day (60% of total)
Debug logs:    25 GB/day (25% of total)
TOTAL:         100 GB/day
```

Cost Calculation:

```
Payment (90d):  10,000 × 90 × 0.0001 × $0.08 = $720/month
Audit (365d):   5,000 × 365 × 0.0001 × $0.08 = $1,460/month
App (35d):      60,000 × 35 × 0.0001 × $0.08 = $1,680/month
Debug (drop):   $0/month
Ingestion:      75,000 × 30 × 0.001 × $0.08 = $180/month

TOTAL:          $4,040/month
```

vs. No Optimization (all 35 days):

```
100 GB/day × 35 days = $3,080/month (but non-compliant!)
```

Key Insight: Sometimes optimization increases cost but ensures compliance. The cost of non-compliance (fines, audits) far exceeds storage costs.

Analyzing Your Bucket Usage

Use these queries to understand your current bucket utilization and plan optimization.

```
// Current bucket distribution for logs
// Shows where your logs are being stored
fetch logs, from: now() - 7d
| summarize {record_count = count()}, by: {dt.system.bucket}
| fieldsAdd daily_avg = round(record_count / 7, decimals: 0)
| sort record_count desc
```

```
// Bucket usage by log source
// Identify which sources are consuming bucket capacity
fetch logs, from: now() - 7d
| summarize {record_count = count()}, by: {dt.system.bucket, log.source}
| sort record_count desc
| limit 30
```

```
// Bucket usage by pipeline
// Shows which pipelines are routing to which buckets
fetch logs, from: now() - 7d
| filter isNotNull(dt.openpipeline.pipelines)
| summarize {record_count = count()}, by: {dt.openpipeline.pipelines,
dt.system.bucket}
| sort record_count desc
```

```
// Bucket usage trend over time
fetch logs, from: now() - 7d
| makeTimeseries {record_count = count()}, by: {dt.system.bucket}, interval:
1d
```

```
// Identify logs that could move to shorter retention buckets
// Debug and trace logs are prime candidates
fetch logs, from: now() - 7d
| filter dt.system.bucket == "default_logs"
| summarize {
    total = count(),
    debug = countIf(loglevel == "DEBUG" OR status == "DEBUG"),
    trace = countIf(loglevel == "TRACE" OR status == "TRACE"),
    info = countIf(loglevel == "INFO" OR status == "INFO")
  }
| fieldsAdd debug_pct = round((toDouble(debug) / toDouble(total)) * 100,
decimals: 2)
| fieldsAdd trace_pct = round((toDouble(trace) / toDouble(total)) * 100,
decimals: 2)
| fieldsAdd optimization_potential = round((toDouble(debug + trace) /
toDouble(total)) * 100, decimals: 2)
```

```
// Check span bucket distribution
// Spans also use buckets
fetch spans, from: now() - 7d
| summarize {span_count = count()}, by: {dt.system.bucket}
| sort span_count desc
```

```
// Estimate storage reduction from tiered retention
// Shows potential savings from routing to different buckets
fetch logs, from: now() - 7d
| summarize {
    total = count(),
    error_warn = countIf(loglevel == "ERROR" OR loglevel == "WARN"),
    info = countIf(loglevel == "INFO"),
    debug_trace = countIf(loglevel == "DEBUG" OR loglevel == "TRACE")
  }
| fieldsAdd tier1_90day = error_warn
| fieldsAdd tier2_35day = info
| fieldsAdd tier3_drop = debug_trace
| fieldsAdd current_cost_units = total * 35
| fieldsAdd optimized_cost_units = (tier1_90day * 90) + (tier2_35day * 35) +
(tier3_drop * 0)
| fieldsAdd savings_pct = round((1.0 - (toDouble(optimized_cost_units) /
toDouble(current_cost_units))) * 100, decimals: 1)
```

Multi-Tier Bucket Strategies ⭐ NEW

Different organizations need different bucket architectures based on their scale, compliance requirements, and cost targets.

3-Tier Strategy (Small to Medium Organizations)

Recommended for: 10-500 GB/day, single or few environments

Tier	Bucket Name	Retention	Use Case	Volume %
Critical	`critical_logs`	90 days	Errors, security, audit	10-15%
Standard	`default_logs`	35 days	WARN, INFO, app logs	60-70%
Ephemeral	`ephemeral_logs`	7 days	DEBUG, health checks	20-30%

Routing Configuration:

```
Route 1 (Critical):
  Condition: loglevel == "ERROR" OR
             contains(log.source, "security") OR
             contains(log.source, "audit") OR
             contains(content, "payment")
  Pipeline:  critical-logs
  Bucket:    critical_logs (90 days)

Route 2 (Ephemeral):
  Condition: loglevel == "DEBUG" OR
             loglevel == "TRACE" OR
             contains(content, "/health")
  Pipeline:  ephemeral-logs
  Bucket:    ephemeral_logs (7 days)

Default Route:
  Pipeline:  standard-logs
  Bucket:    default_logs (35 days)
```

Expected Savings: 25-35% vs. single-bucket approach

5-Tier Strategy (Enterprise Organizations)

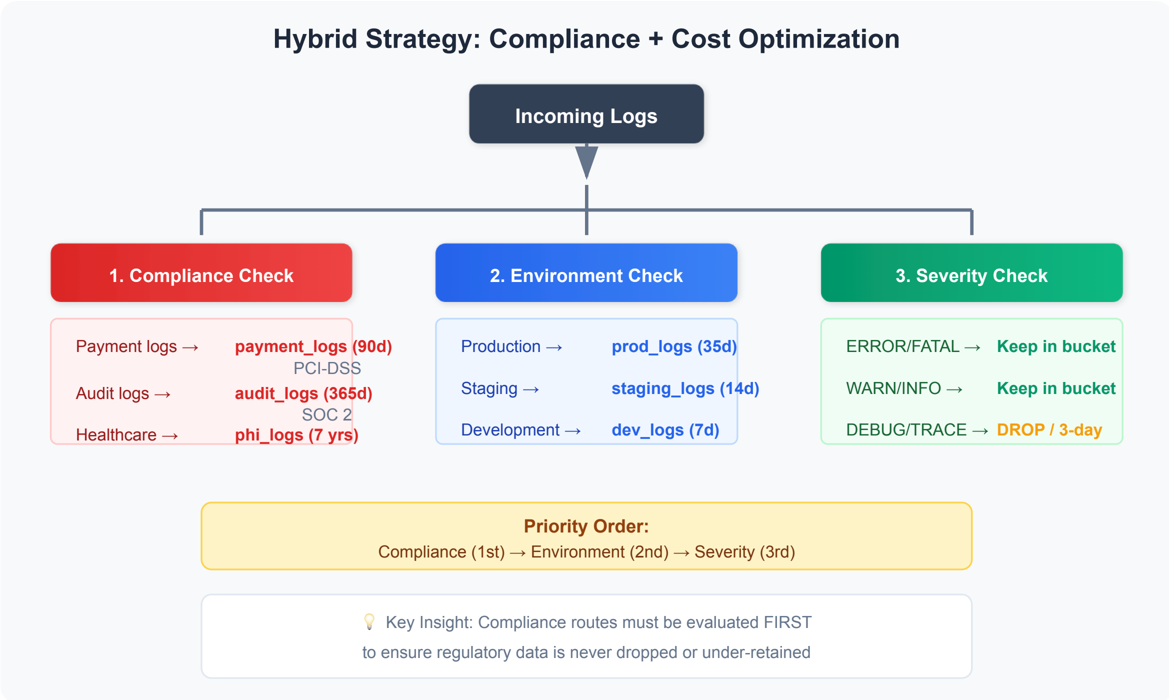
Recommended for: 500+ GB/day, multiple environments, strict compliance

Tier	Bucket Name	Retention	Use Case	Volume %
Compliance	`compliance_logs`	365 days	SOC 2, GDPR, HIPAA audit trails	3-5%
Security	`security_logs`	180 days	Security events, attacks, vulns	2-5%
Production	`prod_logs`	90 days	Production errors, warnings	10-15%
Standard	`default_logs`	35 days	INFO logs, standard operations	40-50%
Development	`dev_logs`	7 days	Dev/staging, DEBUG, testing	30-40%

Expected Savings: 40-60% vs. single-bucket approach

Hybrid Strategy (Compliance + Cost)

Best for: Organizations with compliance requirements AND cost constraints

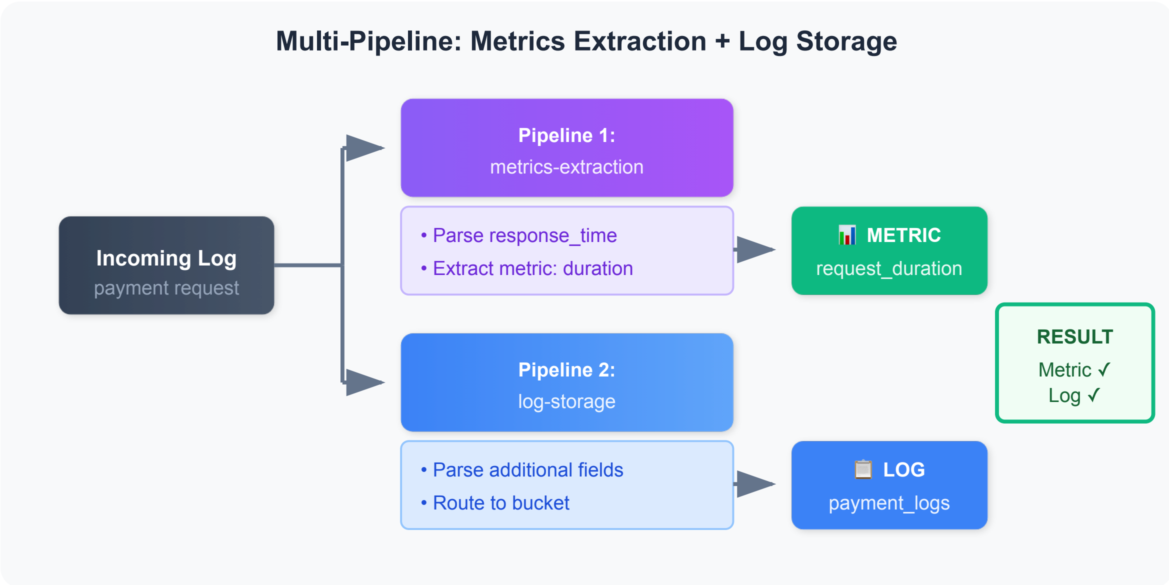


Key Principle: Compliance routes must be FIRST to ensure regulatory data isn't accidentally dropped or under-retained.

Multi-Pipeline Processing

A single record can be processed by multiple pipelines (up to 5). This enables powerful patterns:

Use Case: Extraction + Storage Pipeline



Use Case: Security + Operational Logs

```
Pipeline 1: security-analysis
  - Extract security events
  - Generate security alerts

Pipeline 2: operational-logs
  - Standard parsing
  - Normal storage
```

Configuring Multi-Pipeline Routes

Create overlapping route conditions:

```
Route 1: log.source == "auth-service"  → security-logs
Route 2: log.source == "auth-service"  → app-logs

Both pipelines process the same record
```

> ⚠️ **Limit:** A record can be processed by maximum 5 pipelines. Beyond this, extraction stops.

```
// Identify records processed by multiple pipelines
fetch logs, from: now() - 1h
| filter isNotNull(dt.openpipeline.pipelines)
| filter arraySize(dt.openpipeline.pipelines) &gt; 1
| summarize {multi_pipeline_count = count()}, by: {dt.openpipeline.pipelines}
| sort multi_pipeline_count desc
| limit 20
```

```
// Count of logs by number of pipelines processing them
fetch logs, from: now() - 1h
| filter isNotNull(dt.openpipeline.pipelines)
| fieldsAdd pipeline_count = arraySize(dt.openpipeline.pipelines)
| summarize {record_count = count()}, by: {pipeline_count}
| sort pipeline_count asc
```

Bucket Governance & Access Control ⭐ NEW

Proper bucket governance ensures compliance, security, and operational efficiency.

Access Control Patterns

Pattern 1: Role-Based Bucket Access

Role	Bucket Access	Permissions
Developers	`dev_logs`	Read/Write
	`staging_logs`	Read/Write
	`prod_logs`	Read-only
SRE/Platform	All buckets	Read/Write
Security Team	`security_logs`	Read/Write
	`audit_logs`	Read-only (immutable)
	`compliance_logs`	Read-only (immutable)
Auditors	`audit_logs`	Read-only
	`compliance_logs`	Read-only
Finance	`payment_logs`	Read-only

Pattern 2: Least Privilege by Environment

```
Development Team:
  - FULL access to dev_logs
  - READ access to staging_logs
  - NO access to prod_logs

QA Team:
  - FULL access to staging_logs
  - READ access to dev_logs
  - READ access to prod_logs (for troubleshooting only)

Production SRE:
  - FULL access to prod_logs, error_logs, security_logs
  - READ access to all other buckets
```

Retention Policy Enforcement

Bucket	Min Retention	Max Retention	Enforcement
`dev_logs`	1 day	14 days	Auto-delete after max
`staging_logs`	7 days	30 days	Auto-delete after max
`default_logs`	35 days	35 days	Standard retention
`prod_logs`	90 days	90 days	Protected
`audit_logs`	365 days	2555 days	Immutable after 30d
`payment_logs`	90 days	2555 days	PCI-DSS compliant
`phi_logs`	2555 days	2555 days	HIPAA 7-year rule

Enforcement Methods:
1. **Automated**: Grail automatically purges data after retention period
2. **Immutability**: Compliance buckets can be configured as write-once (after grace period)
3. **Audit Trail**: All retention changes logged in audit_logs

Compliance Audit Trail Requirements

SOC 2 Compliance

Required Logs (365 days minimum):
- Authentication events (login, logout, MFA)
- Authorization changes (role assignments, permission changes)
- Data access (who accessed what, when)
- Configuration changes (pipeline edits, bucket changes)
- Security events (failed auth, suspicious activity)

Bucket Configuration:

```
Bucket: soc2_audit_logs
Retention: 365 days (minimum)
Immutability: Write-once after 30-day grace period
Access: Security team + auditors only
Encryption: At-rest encryption enforced
```

PCI-DSS Compliance

Required Logs (90 days minimum, 12 months recommended):
- Payment transaction logs (amounts MASKED, card numbers REDACTED)
- Access to cardholder data environments
- Authentication to payment systems
- Database queries touching payment data

Bucket Configuration:

```
Bucket: pci_payment_logs
Retention: 365 days (exceeds 90-day minimum)
Processing: MASK credit card numbers, CVV codes
Access: Payment team + security + auditors
Encryption: TLS 1.2+ in transit, AES-256 at rest
Monitoring: Alert on access attempts
```

HIPAA Compliance

Required Logs (6 years minimum, 7 years recommended):
- PHI access logs (who viewed which patient records)
- Authentication to healthcare systems
- Prescription and medical record changes
- Breach detection events

Bucket Configuration:

```
Bucket: hipaa_phi_logs
Retention: 2555 days (7 years)
Processing: MASK patient IDs, SSNs, dates of birth
           HASH medical record numbers
Access: Healthcare IT + compliance officers only
Encryption: FIPS 140-2 compliant
Audit: All bucket access logged to separate audit bucket
```

GDPR Compliance

Special Considerations:
- **Right to Erasure:** Logs with EU citizen PII must be deletable on request
- **Data Minimization:** Only collect necessary personal data
- **Retention Limits:** Cannot retain personal data longer than necessary

Bucket Strategy:

```
Bucket: eu_customer_logs
Retention: 90 days (balancing security and data minimization)
Processing: MASK email addresses, IP addresses
           ADD customer_id_hash (for erasure lookup)
Deletion API: Integrate with GDPR erasure requests
Data Residency: EU-only bucket (if required)
```

Bucket Naming Conventions

Recommended Pattern: `<environment>_<purpose>_logs`

Examples:

```
prod_app_logs
prod_error_logs
prod_security_logs
staging_app_logs
dev_debug_logs
compliance_audit_logs
compliance_soc2_logs
compliance_pci_logs
compliance_hipaa_logs
ephemeral_healthcheck_logs
```

Benefits:
- Clear ownership and purpose
- Easy to identify in DQL queries
- Simplifies access control rules
- Supports automated governance checks

Bucket Lifecycle Management

Quarterly Review Checklist:
- [ ] Review bucket usage and growth trends
- [ ] Validate retention periods still match requirements
- [ ] Check for unused or abandoned buckets
- [ ] Audit access control list changes
- [ ] Verify compliance bucket configurations
- [ ] Analyze query patterns for optimization opportunities
- [ ] Review storage costs vs. budget

Annual Compliance Audit:
- [ ] Verify all compliance buckets meet regulatory retention
- [ ] Confirm masking/redaction rules are working
- [ ] Test data erasure procedures (GDPR)
- [ ] Review audit trail completeness
- [ ] Validate encryption standards
- [ ] Document bucket governance changes

Best Practices

Routing Best Practices

Practice	Description
Specific first	Place specific routes before general ones
Test conditions	Validate matching conditions with sample data
Avoid overlap	Unless intentional multi-pipeline is needed
Document routes	Keep a map of route → pipeline → bucket
Monitor default	High default pipeline volume = missing routes

Bucket Best Practices

Practice	Description
Meaningful names	Use descriptive bucket names
Match retention to value	Critical data = longer retention
Consider query patterns	Frequently queried data in main bucket
Plan for compliance	Audit logs may need 365+ days
Review regularly	Re-evaluate bucket strategy quarterly

Cost Optimization Best Practices

Practice	Impact
Drop DEBUG/TRACE	30-50% volume reduction
Drop health checks	5-20% volume reduction
Short retention for dev	70-80% cost reduction
Extract metrics, drop logs	Metrics persist, logs don't
Sample high-volume noise	Keep visibility, reduce storage

Routing Configuration Examples

Example 1: Environment-Based Routing

Route 1: Production Logs

```
Condition: k8s.namespace.name == "production" OR k8s.namespace.name == "prod"
Pipeline: production-logs
Bucket: production_logs (35 days)
```

Route 2: Development Logs

```
Condition: k8s.namespace.name == "development" OR k8s.namespace.name == "dev"
Pipeline: development-logs
Bucket: dev_logs (7 days)
```

Example 2: Severity-Based Routing

Route 1: Error Logs (high value)

```
Condition: loglevel == "ERROR"
Pipeline: error-logs
Bucket: error_logs (90 days)
Processing: Parse stack traces, enrich with service info
```

Route 2: Debug Logs (low value)

```
Condition: loglevel == "DEBUG" OR loglevel == "TRACE"
Pipeline: debug-logs
Processing: DROP all (or route to 3-day bucket)
```

Example 3: Compliance Routing

Route 1: Audit Logs

```
Condition: contains(log.source, "audit") OR contains(content,
"authentication")
Pipeline: audit-logs
Bucket: audit_logs (365 days)
Processing: Mask PII, add compliance tags
```

Next Steps

Now that you understand routing and buckets, continue with:

Notebook	Focus Area
OPMIG-06	Processing, Parsing & Transformation
OPMIG-07	Metric & Event Extraction
OPMIG-08	Security, Masking & Compliance
OPMIG-09	Troubleshooting & Validation

References

- OpenPipeline Data Flow
- Grail Buckets
- OpenPipeline Limits

Last Updated: December 12, 2025

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